

3643. Flip Square Submatrix Vertically

Difficulty: Easy

Link: <https://leetcode.com/problems/flip-square-submatrix-vertically/description/?envType=daily-question&envId=2026-03-21>

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Problem:

You are given an $m \times n$ integer matrix `grid`, and three integers `x`, `y`, and `k`.

The integers `x` and `y` represent the row and column indices of the **top-left** corner of a **square** submatrix and the integer `k` represents the size (side length) of the square submatrix.

Your task is to flip the submatrix by reversing the order of its rows vertically.

Return the updated matrix.

Example 1:

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16



1	2	3	4
13	14	15	8
9	10	11	12
5	6	7	16

Input: `grid = [[1,2,3,4],[5,6,7,8],[9,10,11,12],[13,14,15,16]]`, `x = 1`, `y = 0`, `k = 3`

Output: `[[1,2,3,4],[13,14,15,8],[9,10,11,12],[5,6,7,16]]`

Explanation:

The diagram above shows the grid before and after the transformation.

Example 2:

3	4	2	3
2	3	4	2



3	4	4	2
2	3	2	3

Input: `grid = [[3,4,2,3],[2,3,4,2]]`, `x = 0`, `y = 2`, `k = 2`

Output: `[[3,4,4,2],[2,3,2,3]]`

Explanation:

The diagram above shows the grid before and after the transformation.

Constraints:

- `m == grid.length`
- `n == grid[i].length`
- `1 <= m, n <= 50`
- `1 <= grid[i][j] <= 100`
- `0 <= x < m`
- `0 <= y < n`
- `1 <= k <= min(m - x, n - y)`